



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE-3.0

27-06-2021

JEE(Main + Advance) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

Test Type : SCORE-3.0

Test Pattern : JEE (Advanced)

TEST DATE : 27-06-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,C	C,D	A,C	A,D	B,D	A,B	A,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	B	C	D	C	B	A	
SEC-3	Q.	1	2	3	4	5		
	A.	112	213	321	312	213		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,D	A,B,D	B,D	A,B,C,D	A,B,C,D	A,B,D	B,C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	C	D	D	A	D	D	
SEC-3	Q.	1	2	3	4	5		
	A.	312	321	321	213	231		

MATHEMATICS

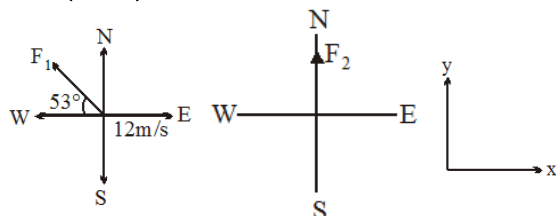
SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,C	B,C	A,D	B,C	C,D	A,B,C	C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	A	C	D	C	D	B	
SEC-3	Q.	1	2	3	4	5		
	A.	231	213	132	221	122		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (A,C)

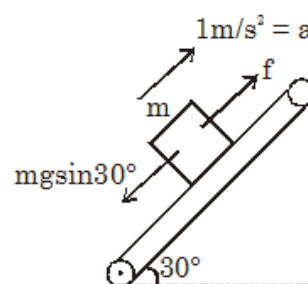
Vel. of air = $v \hat{j}$

$$\vec{v}_{AB} = v \hat{j} - 12 \hat{i}$$

$$\tan 53^\circ = \frac{v}{12} = \frac{4}{3} \quad \therefore v = 16 \text{ m/s}$$

$$v_{AB} = \sqrt{(16)^2 + (12)^2} = 20 \text{ m/s}$$

2. Ans (C,D)



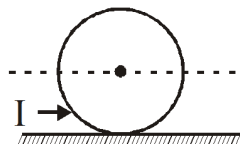
$$N = mg \cos 30^\circ$$

$$f_L = \mu N$$

$$f - mg \sin 30^\circ = ma$$

$$f_L \geq f$$

4. **Ans (A,D)**



$$I = mv_0$$

$$I \times \frac{2R}{3} = \frac{2}{5} mR^2 \omega$$

$$\omega = \frac{5I}{3mR}$$

$$= \vec{v} = \left(\frac{I}{m} + \frac{5I}{3m} \right) \hat{i}$$

Velocity of point of contact is along positive x-axis so friction will be along negative x-axis.

6. **Ans (A,B)**

$$\frac{n\lambda}{2} = L$$

Particles energy in nth state

$$E = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2} = \frac{n^2}{8m\ell^2}$$

7. **Ans (A,D)**

$$200 = 238 \times E - 238 \times 7.6$$

$$\frac{200}{238} + 7.6 = E$$

PHYSICS

SEC-2

1. **Ans (B)**

Time after slipping ceases

$$t = \frac{2v_0 m}{\mu mg \left(1 + \frac{mR^2}{I} \right)}$$

$$I_{\text{Ring}} > I_{\text{Sphere}} > I_{\text{Disk}} > I_{\text{Solid sphere}}$$

$$KE_i = \frac{1}{2} mv_{\text{cm}}^2 + \frac{1}{2} I \omega^2 = \frac{1}{2} mv_0^2 + \frac{1}{2} \frac{I v_0^2}{R^2}$$

$$\Rightarrow KE_i = \frac{1}{2} v_0^2 \left[m + \frac{I}{R^2} \right]$$

$$\Rightarrow KE_f = \frac{1}{2} v^2 \left[m + \frac{I}{R^2} \right]$$

from angular momentum conservation (LAMC)

$$v = \frac{(mR^2 - I) v_0}{(mR^2 + I)}$$

2. **Ans (C)**

Time after slipping ceases

$$t = \frac{2v_0 m}{\mu mg \left(1 + \frac{mR^2}{I} \right)}$$

$$I_{\text{Ring}} > I_{\text{Sphere}} > I_{\text{Disk}} > I_{\text{Solid sphere}}$$

$$KE_i = \frac{1}{2} mv_{\text{cm}}^2 + \frac{1}{2} I \omega^2 = \frac{1}{2} mv_0^2 + \frac{1}{2} \frac{I v_0^2}{R^2}$$

$$\Rightarrow KE_i = \frac{1}{2} v_0^2 \left[m + \frac{I}{R^2} \right]$$

$$\Rightarrow KE_f = \frac{1}{2} v^2 \left[m + \frac{I}{R^2} \right]$$

from angular momentum conservation (LAMC)

$$v = \frac{(mR^2 - I) v_0}{(mR^2 + I)}$$

3. **Ans (D)**

Time after slipping ceases

$$t = \frac{2v_0 m}{\mu mg \left(1 + \frac{mR^2}{I} \right)}$$

$$I_{\text{Ring}} > I_{\text{Sphere}} > I_{\text{Disk}} > I_{\text{Solid sphere}}$$

$$KE_i = \frac{1}{2} mv_{\text{cm}}^2 + \frac{1}{2} I \omega^2 = \frac{1}{2} mv_0^2 + \frac{1}{2} \frac{I v_0^2}{R^2}$$

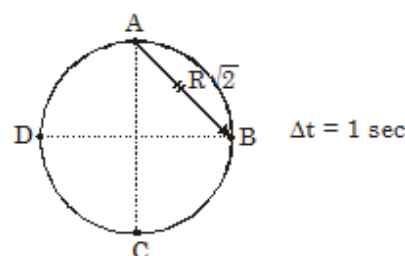
$$\Rightarrow KE_i = \frac{1}{2} v_0^2 \left[m + \frac{I}{R^2} \right]$$

$$\Rightarrow KE_f = \frac{1}{2} v^2 \left[m + \frac{I}{R^2} \right]$$

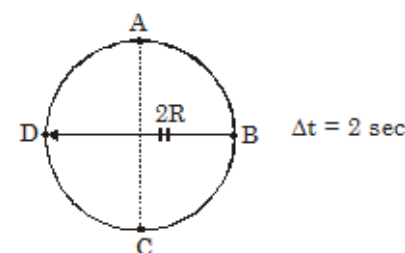
from angular momentum conservation (LAMC)

$$v = \frac{(mR^2 - I) v_0}{(mR^2 + I)}$$

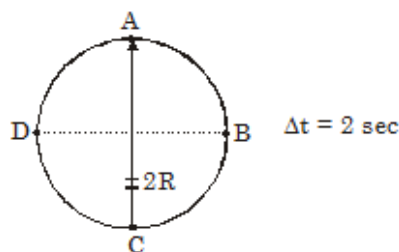
4. **Ans (C)**



5. **Ans (B)**



6. Ans (A)



PHYSICS

SEC-3

3. Ans (321)

(A) Stopping potential in electron volts

$$= h\nu - \phi = 12 - 4 = 8.$$

(B) $\left(\frac{Z_1 - 1}{Z_2 - 1}\right)^2 = \frac{\lambda_2}{\lambda_1} = \left(\frac{85}{81}\right)^2$. Therefore

$$Z_1 = 86 \text{ and } Z_2 = 82$$

(C) Half life time of radioactive material is 4 min. For 80 gm to reduce to 20gm, two half life times are required.

(D) The binding energy per nucleon for helium in MeV is approximately $\frac{0.0302 \times 930}{4} \approx 7$

4. Ans (312)

In x-direction

$$u_x = +8$$

$$a_x = -2$$

$$v_x = (8 - 2t)\hat{i}$$

(P) $v_x = 0$, at $t = 4$ $v_y = 0$, at $t = 4$ sec

(Q) $x = \frac{u_x^2}{2a_x} = 16$ m at $t = 4$ sec

(R) at $t = 0$ & $t = 8$ sec

$$|\vec{v}_{rel}| = 8\sqrt{2} \text{ m/s}$$

In y-direction

$$u_y = +8 \text{ m/s}$$

$$a_y = -2 \text{ m/s}^2$$

$$v_y = (8 - 2t)\hat{j}$$

CHEMISTRY

SEC-1

2. Ans (A,B,D)

(i) Important ingredients present in portland cement are dicalcium silicate (Ca_2SiO_4)26%, tricalcium (Ca_3SiO_5)51% and tricalcium aluminate ($\text{Ca}_3\text{Al}_2\text{O}_6$)11%

(ii) Be & Mg does not produce blue colour in liquid ammonia..

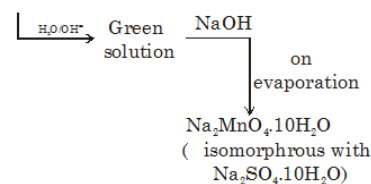
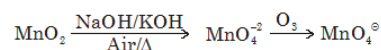
3. Ans (B,D)

(i) $\text{Na}_2\text{Cr}_2\text{O}_7$ is deliquescent & $\text{K}_2\text{Cr}_2\text{O}_7$ is not a deliquescent

(ii) In CrO_4^{2-} & $\text{Cr}_2\text{O}_7^{2-}$ geometry around Cr is tetrahedral

(iii) CrO_4^{2-} & $\text{Cr}_2\text{O}_7^{2-}$ are yellow & orange respectively due to charge transfer

5. Ans (A,B,C,D)



CHEMISTRY

SEC-3

1. Ans (312)

$$w \text{ for (1)} = -1(10 - 5) = -5 \text{ atm litre}$$

$$w \text{ for (1)} = -2(15) = -30 \text{ atm litre}$$

$$w \text{ for (1)} = -1(1 \times 10 \times \ln 2) = -10 \times \ln 2 = -6.93$$

2. Ans (321)

$$(A) \Delta S > 0; \Delta H_{\text{reaction}} = +50 \text{ kJ} > 0$$

$$\Delta G = 50 - T \times \Delta S$$

So T should be high for $\Delta G < 0$

$$(B) \Delta S < 0; \Delta H_{\text{reaction}} = -90 \text{ kJ} < 0$$

$$\Delta G = -90 - T \times \Delta S$$

So T should be low for $\Delta G < 0$

$$B - 2$$

$$(C) \Delta S > 0; \Delta H = -10 < 0$$

$$\Delta G = -10 - T \times \Delta S$$

$$\Delta G < 0 \text{ at all temperature}$$

$$C - 1$$

MATHEMATICS

SEC-1

1. **Ans (B,C)**

$$M = \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}$$

$$M^2 = \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 9 \\ 0 & 4 \end{pmatrix}$$

$$I + C_1 M + C_2 M^2 = \begin{pmatrix} 1 + C_1 + C_2 & 3C_1 + 9C_2 \\ 0 & 1 + 2C_1 + 4C_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

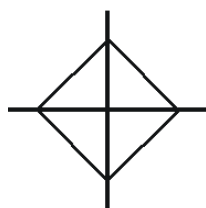
$$C_1 + C_2 = -1 \quad 3C_1 + 9C_2 = 0$$

$$C_1 = -\frac{3}{2} \quad C_2 = \frac{1}{2}$$

2. **Ans (B,C)**

expand and comment

3. **Ans (A,D)**



$$A = \frac{1}{2} \times 6 \times 4 = 12$$

4. **Ans (B,C)**

$$\lim_{x \rightarrow 0^+} x^{2\alpha+1} \ln x = 0$$

$$2\alpha + 1 > 0$$

$$\alpha > -\frac{1}{2}$$

5. **Ans (C,D)**

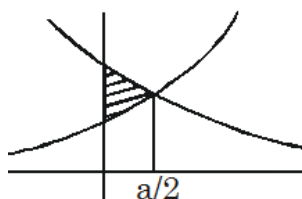
$$AB = \begin{bmatrix} -x & -y & 7z \\ 0 & y & 14z \\ x & -y & 7z \end{bmatrix}$$

$$B^T AB = \begin{pmatrix} 2x^2 & 0 & 0 \\ 0 & 3y^2 & 0 \\ 0 & 0 & 42z^2 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 42 \end{pmatrix}$$

$$2x^2 = 8, \quad 3y^2 = 27, \quad 42z^2 = 42$$

$$x = \pm 2, \quad y = \pm 3, \quad z = \pm 1$$

6. **Ans (A,B,C)**

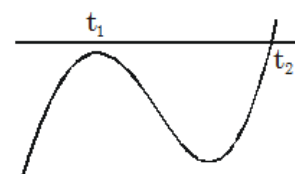


$$A = \int_0^{a/2} (e^{a/2-x} - e^x) dx = (e^{a/2} - 1)^2$$

$$\lim_{a \rightarrow 0} \frac{e^{a/2} - 1}{4 \frac{a^2}{4}} = \frac{1}{4}$$

$$n = 4$$

7. **Ans (C,D)**



$$\left. \frac{dy}{dx} \right|_{t_1} = - \left. \frac{1}{dx/dx} \right|_{t_2} = \frac{2t_2^3 - 2t_1^3}{3(t_2^2 - t_1^2)}$$

$$\frac{6t_1^2}{6t_1} = - \frac{6t_2}{6t_2^2} = \frac{2}{3} \frac{(t_2^2 + t_1^2 + t_1 t_2)}{t_2 + t_1}$$

$$t_1 = -\frac{1}{t_2} = \frac{2}{3} \frac{t_2^2 + t_1^2 + t_1 t_2}{t_1 + t_2}$$

$$t_1 = \pm \sqrt{2}$$

Straight lines are

$$y = -\sqrt{2}(x - 2)$$

$$y = \sqrt{2}(x - 2)$$

MATHEMATICS

SEC-2

1. Ans (A)

(I) $y = \sin x$ $m_1 = y' = \cos x = \frac{1}{\sqrt{2}}$

$y = \cos x$ $m_2 = y' = -\sin x = -\frac{1}{\sqrt{2}}$

$\sin x = \cos x$

$\tan x = 1$

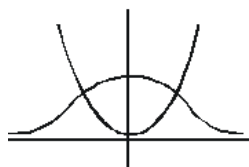
$x = \frac{\pi}{4}, \frac{5\pi}{4}$

$\tan \theta = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right| = \left| \frac{\frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right)}{1 - \frac{1}{2}} \right|$

$\theta = \tan^{-1} 2\sqrt{2}$

(II) $x^2 = 4y$ $y = \frac{8}{x^2 + 4}$

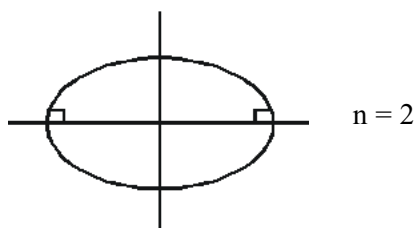
POI $y = \frac{8}{4y + 4}$



$y^2 + y = 8$

$y = \frac{-1 \pm \sqrt{1 + 32}}{2}$ $2 = n$

(III) $\frac{x^2}{18} + \frac{y^2}{8} = 1$ and $y = 0$



$n = 2$

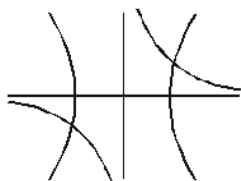
$\theta = \frac{\pi}{2}$

(IV) $xy = 1$ $x^2 - y^2 = 5$

$xy' + y = 0$ $2x - 2yy' = 0$

$m_1 = y' = -y/x$ $m_2 = y' = \frac{x}{y}$

$m_1 m_2 = -1$ $\theta = \frac{\pi}{2}$



$n = 2$

2. Ans (C)

(I) $y = \sin x$ $m_1 = y' = \cos x = \frac{1}{\sqrt{2}}$

$y = \cos x$ $m_2 = y' = -\sin x = -\frac{1}{\sqrt{2}}$

$\sin x = \cos x$

$\tan x = 1$

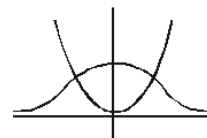
$x = \frac{\pi}{4}, \frac{5\pi}{4}$

$\tan \theta = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right| = \left| \frac{\frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}}\right)}{1 - \frac{1}{2}} \right|$

$\theta = \tan^{-1} 2\sqrt{2}$

(II) $x^2 = 4y$ $y = \frac{8}{x^2 + 4}$

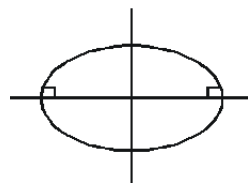
POI $y = \frac{8}{4y + 4}$



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$n = 2$

$\theta = \frac{\pi}{2}$

(IV) $xy = 1$ $x^2 - y^2 = 5$

$xy' + y = 0$ $2x - 2yy' = 0$

$m_1 = y' = -y/x$ $m_2 = y' = \frac{x}{y}$

$m_1 m_2 = -1$ $\theta = \frac{\pi}{2}$



$n = 2$

3. Ans (D)

$$(I) \quad y = \sin x \quad m_1 = y' = \cos x = \frac{1}{\sqrt{2}}$$

$$y = \cos x \quad m_2 = y' = -\sin x = -\frac{1}{\sqrt{2}}$$

$$\sin x = \cos x$$

$$\tan x = 1$$

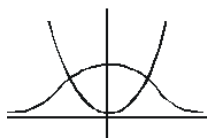
$$x = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\tan \theta = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right| = \left| \frac{\frac{\sqrt{2}}{1} - \frac{1}{\sqrt{2}}}{1 - \frac{1}{2}} \right|$$

$$\theta = \tan^{-1} 2\sqrt{2}$$

$$(II) \quad x^2 = 4y \quad y = \frac{8}{x^2 + 4}$$

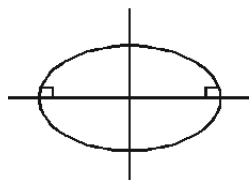
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$$y^2 + y = 8$$

$$y = \frac{-1 \pm \sqrt{1 + 32}}{2} \quad 2 = n$$

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$$n = 2$$

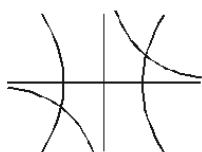
$$\theta = \frac{\pi}{2}$$

$$(IV) \quad xy = 1 \quad x^2 - y^2 = 5$$

$$xy' + y = 0 \quad 2x - 2yy' = 0$$

$$m_1 = y' = -y/x \quad m_2 = y' = \frac{x}{y}$$

$$m_1 m_2 = -1 \quad \theta = \frac{\pi}{2}$$



$$n = 2$$

4. Ans (C)

$$(I) \quad y = \sin x + \cos x \quad (II) \quad y = \sin 2x$$

$$y' = \cos x - \sin x \quad y' = 2\cos 2x$$

$$y \uparrow \left(0, \frac{\pi}{4}\right) \quad y \uparrow \left(0, \frac{\pi}{4}\right)$$

$$y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \quad y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$(III) \quad y = \sin^5 x + \cos^5 x - 1$$

$$y' = 5\sin^4 x \cos x - 5\cos^4 x \sin x$$

$$= 5\sin x \cos x (\sin^3 x - \cos^3 x)$$

$$y \uparrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$y \downarrow \left(0, \frac{\pi}{4}\right)$$

$$y = x^e \quad y' = x^x(1 + \ln x)$$

$$y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$$

$$y \downarrow \left(0, \frac{1}{e}\right)$$

5. Ans (D)

$$(I) \quad y = \sin x + \cos x \quad (II) \quad y = \sin 2x$$

$$y' = \cos x - \sin x \quad y' = 2\cos 2x$$

$$y \uparrow \left(0, \frac{\pi}{4}\right) \quad y \uparrow \left(0, \frac{\pi}{4}\right)$$

$$y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \quad y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$(III) \quad y = \sin^5 x + \cos^5 x - 1$$

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$$y = x^e \quad y' = x^x(1 + \ln x)$$

$$y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$$

$$y \downarrow \left(0, \frac{1}{e}\right)$$

6. Ans (B)

(I) $y = \sin x + \cos x$ (II) $y = \sin 2x$
 $y' = \cos x - \sin x$ $y' = 2\cos 2x$
 $y \uparrow \left(0, \frac{\pi}{4}\right)$ $y \uparrow \left(0, \frac{\pi}{4}\right)$
 $y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ $y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(III) $y = \sin^5 x + \cos^5 x - 1$
 $y' = 5\sin^4 x \cos x - 5\cos^4 x \sin x$
 $= 5\sin x \cos x (\sin^3 x - \cos^3 x)$
 $y \uparrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
 $y \downarrow \left(0, \frac{\pi}{4}\right)$
 $y = x^e$ $y' = x^x(1 + \ln x)$
 $y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$
 $y \downarrow \left(0, \frac{1}{e}\right)$

MATHEMATICS

SEC-3

1. Ans (231)

(P) $L = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{\frac{T}{2\pi} 2\pi} (\sin x + \sin ax)^2 dx$
 $= \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_0^{2\pi} (\sin x + \sin ax)^2 dx$
 $a = 0 \quad L = \frac{1}{2\pi} \int_0^{2\pi} (\sin x + 0)^2 dx$
 $= \frac{1}{2\pi} \int_0^{2\pi} \frac{1 + \cos 2x}{2} dx$
 $= \frac{1}{4\pi} 2\pi = \frac{1}{2}$
 $a = 1 \quad L = \lim_{T \rightarrow \infty} \frac{1}{2\pi} \int_0^{2\pi} (2 \sin x)^2 dx$
 $= \frac{4}{2\pi} \frac{1}{2} 2\pi = 2$
 $a = -1 \quad L = 0$

2. Ans (213)

(P) $I = \int_{-1}^1 \frac{1+x^3 - \sqrt{1+x^6}}{1+x^6 + 2x^3 - 1 - x^6} dx$
 $= \int_{-1}^1 \frac{1}{x^3} + \frac{1}{2} \int_{-1}^1 dx - \int_{-1}^1 \frac{\sqrt{1+x^6}}{2x^3} dx$
 $= 1$

(Q) $f(x) = \frac{1}{x} - \frac{1}{x^2}$
 $f'(x) = -\frac{1}{x^2} + \frac{2}{x^3}$
 $f''(x) = +\frac{2}{x^3} - \frac{6}{x^4} = 0$
 $x = 3$

(R) $I = \int_1^e \frac{1}{x(\ln x + 1)} dx$
 $\ln x = t$
 $\frac{1}{x} dx = dt$
 $= \int_0^1 \frac{dt}{t+1} = (\ln t + 1)_0^1$
 $= \ln 2$

3. Ans (132)

(P) expand
 (Q) $|C| = |B| |A| |B|^{-1} |B|^{-1} |A| |B|$
 $= |A|^2 = 4^2 = 16$
 (R) $A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = I + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = I + B$
 $B^2 = I$
 $A^4 = (I + B)^4$
 $= {}^4C_0 I + {}^4C_1 B + {}^4C_2 B^2 + {}^4C_3 B^3 + {}^4C_4 B^4$
 $= -4I$

4. Ans (221)

(P) $I = \int_0^1 \ln \frac{1-x}{x} dx$ Property (5) and

add $I = 0$

(Q)
$$\int_0^{10\pi} \lim_{x \rightarrow y} \left(\frac{\sin x - \sin y}{x - y} \right) dx = \int_0^{10\pi} \cos y dy$$

 $= 0$

(R) $I = \sum \frac{1^2}{1^3 + n^3} = \int_0^2 \frac{x^2}{x^3 + 1} dx$

$x^3 + 1 = t$

$= \int_1^9 \frac{\frac{1}{3} dt}{t} = \frac{1}{3} (\ln t)_1^9$

$= \frac{1}{3} \ln 9$

5. Ans (122)

(P) $\sqrt{x^2 + y^2} = \frac{y\sqrt{1+m^2}}{m}$
 $x^2 m^2 + m^2 y^2 = y^2 + m^2 y^2$

$\frac{dy}{dx} = \pm \frac{y}{x}$

$\int \frac{dy}{y} = \pm \int \frac{dx}{x}$

$\ln y = \pm \ln x + \ln c$

$xy = k$ $y = k'x$ straight line

(Q) tangent $y - g = m(x - x)$

$B(0, y - mx)$ $P(x, y)$

$PB = OB$

$\sqrt{x^2 + m^2 x^2} = |y - mx|$

$x^2 + m^2 x^2 = y^2 + m^2 x^2 - 2mxy$

$2xy \frac{dy}{dx} = y^2 - x^2$

$\frac{2xy dy - y^2 dx^2}{x^2} = \frac{-x^2 dx}{x^2}$

$d \left(\frac{y^2}{x} \right) = - \int dx$

$\frac{y^2}{x} = -x + c$

$y^2 + x^2 + cx = 0$ circle

(R) $\sqrt{x^2 + y^2} = y_1 \sqrt{1 + m^2}$

$x^2 + y^2 = y^2 + y^2 m^2$

$\frac{dy}{dx} = \pm \frac{x}{y}$

$\frac{y^2}{2} = \pm \frac{x^2}{2} + c$

circle